TRIBIKRAM DHAR

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WORK EXPERIENCE

Researcher, Tata Consultancy Services (TCS) Research — 2023 – present
RoboVerse CTO1.0, TCS Research — *Bengaluru, India*

- Developed and maintained testbeds for iiwa14 and UR10 robots for protype testing and research.
- Designed and optimised motion planning libraries.
- Optimised voxel-based perception modules and distance field modules.
- Built an end-to-end VLM pipeline for a natural-language-to-manipulation project.
- Published research studies in international and national conferences.
- Supervised undergraduate research interns on projects.

EDUCATION

Jadavpur University — Kolkata, India
Bachelor of Engineering in Electrical Engineering — *2019-2023*

- Graduated with First Class Distinction; CGPA: 8.65/10

RESEARCH INTERESTS

Robotic Manipulation, Robot Learning, Generative Control Policies, Reinforcement Learning, Deep Learning for Robotics

RESEARCH PROJECTS

Autonomous Retail Shelf Restocking with Physics-Based Sampling MPC Video (google drive) June 2026 – present

- Built a crate-to-shelf restocking pipeline on a KUKA iiwa14 combining open-set perception, RANSAC-based grasp pose estimation, and planogram slot allocation.
- Designed a single 70 Hz joint-space MPPI controller for reach, narrow-shelf insertion, and retraction.
- Checked collisions for sampled rollouts in GPU-parallel Newton physics against a digital twin of the shelf, including the held item.
- Achieved 92.7% placement success over 124 hardware placements with 14.8 mm mean placement error.

Robot Skill Library using Hybrid RL-MPC Framework July 2025 – February 2026

- Trained low-level configuration-space controllers for iiwa14 using PPO and SAC in parallel environments.
- Implemented sim-to-real transfer of the low-level control policies on iiwa14.
- Designed non-linear MPCs for task-space control of the end-effector for manipulation.
- Designed and implemented RL and MPC based policies for autonomous inventory item manipulation.

Safe Motion Planning for Human-Obstacle Avoidance | Video (google drive) January 2026 – June 2026

- Designed an optimal controller policy for a serial manipulator that avoids the workshop table and reacts to human intrusion in the robot's path.
- Integrated signed distance fields into the cost formulation for differentiable cost computation.
- Added an articulated human cost model using live keypoint extraction from an RGBD stream.
- Designed an MPPI-based policy for goal-reaching, obstacle-avoidance, and human-avoidance tasks.
- Tested across diverse scenarios in simulation and on hardware, achieving approximately 95% success rates for goal-reaching in scenes with cartons and shelves.

Task and Motion Planning via Instruction-to-Subtask Decomposition | Video (google drive) December 2024

- Developed a reusable C++ motion-primitive library for a UR10 with an RG2 gripper, driving reliable bin-picking and grasping.
- Fine-tuned a MiniGPT-4-based vision-language planner to decompose free-form language instructions into executable long-horizon subtask sequences.
- Engineered low-latency ROS1 services that decoupled planning from execution, making the pipeline modular and robot-agnostic.
- Presented the work as a workshop paper at ICC 2025 Workshop.

Technical Skills

Programming Languages: C++, Python
Frameworks and Libraries: PyTorch, NumPy, CasADi, OpenCV, Eigen, Matplotlib, ROS 1 Noetic, ROS 2 Kilted
Robot Hardware: UR10 CB3, Kuka LBR iiwa 14, Intel Realsense, OnRobot FT sensor
Simulation engines: NVIDIA Isaac (Lab + Sim), Isaac Gym, Mujoco, Gymnasium, Newton
Development Tools and Environments: Git, CMake, Linux (Ubuntu)

Selected Publications

- **T.Dhar**, P. N. Jayantkumar, R.Dey, S.Dutta, "*Sample-Based MPC for Safe Motion Planning in Human-Shared Manipulator Workspaces*", Indian Control Conference, 2027. (**Accepted**)
- R.Dey, **T.Dhar**, S.Dutta, "*Aligning Heirarchical Vision Transformers as Multi-Scale Dense Feature Extractors*", International Joint Conference on Neural Networks, 2026. IEEEXplore
- **T.Dhar**, G.Adhikari, S.Chaudhuri, "*An Improved Classification of Chest X-ray Images Using Adaptive Activation Function*", International Conference on Electronics, Materials Engineering and Nano-Technology, 2021. IEEEXplore

Workshop Paper

- **T. Dhar**, R. Dey, P. N. Jayantkumar, S. Dutta, "*Instruct2Subtask: A Language Parsing Framework for Sequential Robotic Manipulation*", Indian Control Conference 2025 Workshop on Advances in AI-Driven Control and Estimation for Robotics and Autonomous Systems. OpenReview

Preprints

- *Autonomous Retail Restocking via Unified Joint-Space MPPI with Real-Time Physics Rollouts.* (**under Double Blind Review**)